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Semester: III	Year : AY 2026-27
Subject: Programming with Java	
ASSIGNMENT No : 5	
TITLE : Collection Classes and StringBuffer	

Problem Statement

Develop a Java application demonstrating the use of ArrayList, Vector, and StringBuffer for dynamic data storage, management, and string manipulation.

Learning Objectives

To implement dynamic data storage and string manipulation using ArrayList, Vector, and StringBuffer.

Programme

1.Question1-1.

Create a To-Do List application using ArrayList to store tasks and StringBuffer to display tasks

```
assignment5 > J Todolist.java > Todolist > main(String[] args)
1  import java.util.ArrayList;
2
3  public class Todolist {
    Run main | Debug main
4      public static void main(String[] args) {
5          ArrayList<String> tasks = new ArrayList<>();
6
7          tasks.add("Going to collage");
8          tasks.add("Washing Cloths");
9          tasks.add("Study Java");
10         tasks.add("Take a nap");
11
12
13
14         tasks.remove("Take a nap");
15
16         StringBuffer sb = new StringBuffer();
17         sb.append("To Do List:\n");
18         for (int i = 0; i < tasks.size(); i++) {
19             sb.append(tasks.get(i)).append("\n");
20         }
21
22         System.out.println(sb);
23     }
24 }
```

Problems 1 Output Debug Console Terminal Ports

Output-

```
PS C:\Users\dhruv\OneDrive\Pictures\Documents\Desktop\Study\sem 3\JAVA\assignment5> javac Todolist.java
PS C:\Users\dhruv\OneDrive\Pictures\Documents\Desktop\Study\sem 3\JAVA\assignment5> java Todolist
To Do List:
Going to collage
Washing Cloths
Study Java
PS C:\Users\dhruv\OneDrive\Pictures\Documents\Desktop\Study\sem 3\JAVA\assignment5>
```

Question2-

Create a Student Course Registration System using ArrayList to store the list of courses registered by a student and StringBuffer to generate and display the registered course list. The application should allow users to add, remove, and view registered courses

```
assignment5 > J CourseReg.java > % CourseReg > main(String[] args)
1  import java.util.ArrayList;
2
3  public class CourseReg {
    Run main | Debug main
4  public static void main(String[] args) {
5      ArrayList<String> courses = new ArrayList<>();
6
7      courses.add("Programming With Java");
8      courses.add("DBMS");
9      courses.add("OS");
10     courses.add("Data Structure");
11     courses.add("Mathematics");
12
13
14     courses.remove("Programing With Java");
15     courses.remove("OS");
16
17     StringBuffer sb = new StringBuffer();
18     sb.append("Registered Courses:\n");
19     for (int i = 0; i < courses.size(); i++) {
20         sb.append(courses.get(i)).append("\n");
21     }
22
23     System.out.println(sb);
24 }
25 }
```

Output-

```
PS C:\Users\dhruv\OneDrive\Pictures\Documents\Desktop\Study\sem 3\JAVA\assignment5> git push
PS C:\Users\dhruv\OneDrive\Pictures\Documents\Desktop\Study\sem 3\JAVA\assignment5> cd "c:\Users\dhruv\OneDrive\Pictures\Documents\Desktop\Study\sem 3\JAVA\assignment5\" ; if ($?) { javac CourseReg.java } ; if ($?) { java CourseReg }
Registered Courses:
Programming With Java
DBMS
Data Structure
Mathematics
PS C:\Users\dhruv\OneDrive\Pictures\Documents\Desktop\Study\sem 3\JAVA\assignment5> 
```

Conclusion-

We learn To implement dynamic data storage and string manipulation through arraylist and stringbuffer.